



A human life-stage physiologically based pharmacokinetic and pharmacodynamic model for chlorpyrifos: Development and validation



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ABSTRACT

Sensitivity to some chemicals in animals and humans are known to vary with age. Age-related changes in sensitivity to chlorpyrifos have been reported in animal models. A life-stage physiologically based pharmacokinetic and pharmacodynamic (PBPK/PD) model was developed to predict disposition of chlorpyrifos and its metabolites, chlorpyrifos-oxon (the ultimate toxicant) and 3,5,6-trichloro-2-pyridinol (TCPy), as well as B-esterase inhibition by chlorpyrifos-oxon in humans. In this model, previously measured age-dependent metabolism of chlorpyrifos and chlorpyrifos-oxon were integrated into age-related descriptions of human anatomy and physiology. The life-stage PBPK/PD model was calibrated and tested against controlled adult human exposure studies. Simulations suggest age-dependent pharmacokinetics and response may exist. At oral doses ≥ 0.6 mg/kg of chlorpyrifos (100- to 1000-fold higher than environmental exposure levels), 6 months old children are predicted to have higher levels of chlorpyrifos-oxon in blood and higher levels of red blood cell cholinesterase inhibition compared to adults from equivalent doses. At lower doses more relevant to environmental exposures, simulations predict that adults will have slightly higher levels of chlorpyrifos-oxon in blood and greater cholinesterase inhibition. This model provides a computational framework for age-comparative simulations that can be utilized to predict chlorpyrifos disposition and biological response over various postnatal life stages.

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1. Introduction

For some chemical exposures including pesticides, children may have different responses than adults (Bruckner, 2000), and thus an increased emphasis has been placed on evaluating effects of chemical exposures in different age groups. Mechanisms of absorption, distribution, metabolism, and excretion all develop at different rates, which could lead to age-dependent discrepancies in both pharmacokinetics and pharmacodynamics (US EPA, 2006; US NRC, 1993). In addition, it has been suggested that variable degrees of increased susceptibility exist in young animals compared to adults from chlorpyrifos exposures (Marty et al., 2012; Pope et al., 1991; Timchalk et al., 2006; Vidair, 2004; Whitney et al., 1995; Zheng et al., 2000).

Chlorpyrifos (CPF) is a broad spectrum organophosphate insecticide used in crop agriculture. Once absorbed in mammals, cytochrome P450 enzymes metabolize chlorpyrifos to undergo either oxidative desulfuration (Fig. 1), forming chlorpyrifos-oxon, or

dearylation (oxidative ester cleavage), forming 3,5,6-trichloro-2-pyridinol (TCPy) and diethylthiophosphate (Kamataki et al., 1976). Chlorpyrifos-oxon is a potent inhibitor of cholinesterases, including acetylcholinesterase (AChE) and butyrylcholinesterase (BuChE), by covalently binding to the active site of these enzymes (Sultatos, 1994). Chlorpyrifos-oxon can be metabolized by hepatic and extrahepatic A-esterases, such as paraoxonase-1 (PON1) (Pond et al., 1998), and tissue B-esterases (e.g. carboxylesterase, CaE) to form TCPy and diethylphosphate (Chanda et al., 1997). Formation of TCPy (by either chlorpyrifos dearylation or chlorpyrifos-oxon hydrolysis) is generally considered to be a detoxification pathway, and TCPy can be further metabolized by phase II enzymes to form various conjugates (Bakke et al., 1976; Nolan et al., 1984). Conjugated and unconjugated TCPy, diethylphosphate, and diethylthiophosphate are excreted in urine (Bakke et al., 1976; Nolan et al., 1984).

Results from studies have suggested increased susceptibility in young animals compared to adults from acute chlorpyrifos exposures, the degree of which is variable with age, dose, and target tissue (Marty et al., 2012; Pope et al., 1991; Timchalk et al., 2006; Vidair, 2004; Whitney et al., 1995; Zheng et al., 2000). In one study, age-dependent sensitivity to cholinesterase inhibition of varying

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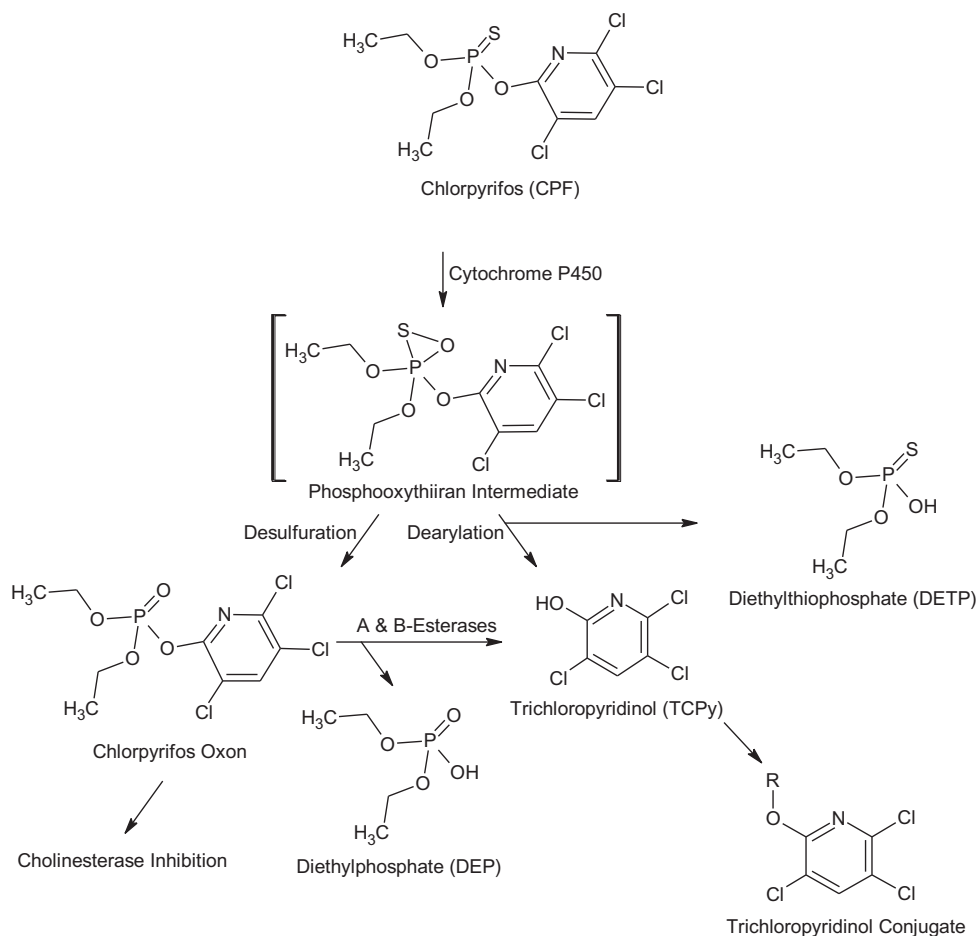


Fig. 1. Enzymatic metabolism of chlorpyrifos (CPF), chlorpyrifos-oxon (CPF-oxon), and 3,5,6-trichloro-2-pyridinol (TCPy).

degrees were demonstrated in plasma, red blood cells, and brain of postnatal day (PND) 5, 12, and 17 rats dosed with 1 and 10 mg/kg chlorpyrifos orally, where PND 5 animals appeared to be more sensitive than adults at both doses, but PND 12 rats were only more sensitive at the higher dose (Timchalk et al., 2006). In agreement with those results, PND 12 rats had increased red blood cell cholinesterase inhibition compared to adults (2 mg/kg chlorpyrifos, orally), but demonstrated similar plasma cholinesterase activities and equivalent no adverse effect levels (NOEL; 0.5 mg/kg), indicating negligible age-dependent effects at lower doses for those age groups (Marty et al., 2012). A benchmark dose analysis (BMD) on that study (Marty et al., 2012) and meta-analysis of other studies recommends equivalent BMD₁₀ values for both children and adults after single acute (1.7 mg/kg/day) and repeated (0.67 mg/kg/day) doses (Reiss et al., 2012).

Several age-related factors could contribute to age-dependent susceptibility including levels of metabolism and variable growth rates of organs. Age-related increases in both chlorpyrifos desulfuration and dearylation pathways on a liver mass basis (Atterberry et al., 1997), and age-dependent changes in chlorpyrifos-oxon hydrolysis (Karanth and Pope, 2000) have all been reported in rats. In humans, hepatic chlorpyrifos desulfuration, chlorpyrifos dearylation, and chlorpyrifos-oxon hydrolysis have not demonstrated any age-dependent relationships on a microsomal protein basis; however, when scaling those values based on liver volume, age-dependent differences existed (Smith et al., 2011). Chlorpyrifos-oxon hydrolysis in plasma demonstrated an age-dependent increase on a volume of plasma basis in humans, with mean chlorpyrifos-oxon metabolism increasing over 3.5

times from <6 months in age to adulthood (Smith et al., 2011). Differing rates of organ growth can also affect distribution of chemicals. For example, relative fat levels in children are lower than in adults, and conversely, relative brain size is higher in children than in adults. Since approximately 11% of brain tissue is lipid compounds (Poulin and Krishnan, 1995), high brain to fat tissue ratios (as found in children) could augment partitioning of lipophilic chemical compounds to brains of children compared to that of adults.

Physiologically based pharmacokinetic and pharmacodynamic (PBPK/PD) models provide a computational framework to quantitatively incorporate age-dependent changes in physiology and mechanisms of chemical disposition to predict net effects on target tissue dosimetry and biological response. Several models have been developed to assess age-dependent risk including caffeine (Ginsberg et al., 2004), theophylline (Ginsberg et al., 2004), perchlorate (Clewett et al., 2007), bisphenol A (Willhite et al., 2008), and chloroform (Liao et al., 2007). PBPK/PD models have been developed for chlorpyrifos including models for 'typical' rats and humans (Timchalk et al., 2002), chemical mixtures (Timchalk and Poet, 2008), gestational exposure (Lowe et al., 2009), and age-dependency in rats (Timchalk et al., 2007b). Hinderliter et al., (2011) developed estimates of response for two specific age groups (3 and 30 years) linked to dietary exposure, and Foxenberg et al., (2011) made predictions of two age groups (1 and 19 years) exposed to two organophosphates (parathion and chlorpyrifos) based on extrapolating cytochrome P450-specific activities; however, a general age-dependent of chlorpyrifos in humans has not been developed.

The purpose of this study was to quantitatively integrate age-dependent parameters of physiology as well as chlorpyrifos pharmacokinetics and pharmacodynamics into a life-stage model for humans. Mathematical descriptions of age-dependent changes in body weight, organ volumes, and metabolism from the literature and experimental studies were incorporated into the model which was tested against controlled adult human chlorpyrifos exposure studies. This model will be useful for predicting chlorpyrifos disposition and biological response across life-stages.

2. Materials and methods

2.1. Chemicals and samples

Chlorpyrifos-oxon was kindly provided by the Dow Chemical Company (Midland, MI, USA). Mixed human plasma was obtained from Bioreclamation, Inc. (Liverpool, NY, USA). Other general laboratory chemicals were purchased from Sigma–Aldrich (St. Louis, MO, USA) and were reagent grade or better.

2.2. Human chlorpyrifos pharmacokinetic studies

There are two controlled human chlorpyrifos exposure studies available for model development that have been evaluated for both scientific and ethical considerations by the US Environmental Protection Agency Human Studies Review Board. In the first study, human volunteers were administered 0.5 mg/kg chlorpyrifos orally by applying chlorpyrifos (in solvent that was subsequently evaporated off) to the surface of a lactose tablet (Nolan et al., 1984). Endpoints measured included TCPy levels in blood and urine as well as chlorpyrifos levels in blood and plasma cholinesterase activity. In the second study, volunteers were administered 0.5, 1, or 2 mg/kg chlorpyrifos (neat) orally in a capsule (Kisicki et al., 1999; Timchalk et al., 2002). Measured endpoints were chlorpyrifos levels in blood, TCPy levels in blood and urine, and cholinesterase activity in red blood cells.

2.3. Model structure

A life-stage PBPK/PD model was developed to describe the absorption, distribution, metabolism, and excretion of CPF. The PBPK/PD model structure was based on the original model developed for diisopropylfluorophosphate (DFP) (Gearhart et al., 1990), which has subsequently been adapted for chlorpyrifos (Timchalk et al., 2002) and further modified for age-dependent changes in rats (Timchalk et al., 2007b) (Fig. 2). The model describes the time-course disposition of chlorpyrifos, chlorpyrifos-oxon, and TCPy as well as inhibition of several B-esterases by chlorpyrifos-oxon including acetylcholinesterase, butyrylcholinesterase, and carboxylesterase.

Body weight and compartmental volumes for the life-stage PBPK/PD model were scaled using an approach proposed by Luecke et al. (2007) and Young et al. (2009). Using this approach, body weight was scaled by age using a modified Gompertz growth function (Eq. 1A for t_0 to t_2 and 1B for t_2 to ∞) that can be parameterized by gender or species, where body weight (BWT [kg]), and W_x , α_x , R_x , and β_x are parameters that vary in steps by time denoted as t_x . These generalized equations were flexible enough to describe growth in a number of genders and species (for extrapolation) and specific enough to describe growth phenomena such as adolescence growth spurts (Luecke et al., 2007).

$$\text{BWT}(t)_{t_x:t_{x+1}} = W_x \times e^{\frac{R_x}{e^{2R_x} \times (1 - e^{-2\alpha_x t_x})}} + \beta_x(t - t_x) \quad (1A)$$

$$\text{BWT}(t)_{t_x:t_{x+1}} = W_x \times e^{\frac{R_x}{e^{2R_x} \times (1 - e^{-2\alpha_x (t - t_x)})}} + \beta_x(t - t_x) \quad (1B)$$

The same compartments were used in the life-stage model that were used in the original PBPK/PD model (Timchalk et al., 2002) and included brain, blood, diaphragm, fat, liver, rapidly perfused, and slowly perfused compartments. The rapidly perfused compartment consisted of a summation of kidney, spleen, lung, gastrointestinal tract (GI), and pancreas volumes; and the slowly perfused compartment consisted of a summation of muscle, skin, bone marrow, and non-fat adipose tissue volumes.

The life-stage PBPK/PD model incorporated both adipose and lipid compartments descriptions in the model. Adipose tissue is defined as all tissue associated with adipose tissue (connective and lipid tissue), while the fat compartment is lipid content only. The non-fat adipose tissue (connective tissue, adipose compartment subtract the fat compartment) was included in the slow compartment. This helps to balance out the total volume of distribution and more accurately represent actual lipid levels in relation to adipose levels. The International Commission on Radiological Protection (ICRP) Publication 89 (2002) describes adipose tissue as a combination of fat and connective tissue, and reference fat content of adipose tissue is 40% at birth and increases to ~80% in adults. Thus it was important to account for these differences, as failure to do so could lead to age-related inaccurate compartment volume estimates for fat or fat partition coefficients (if one assumes a constant lipid content of adipose tissue). These potential inaccuracies could have substantial implication on overall pharmacokinetics for lipophilic compounds like chlorpyrifos.

Compartmental volumes (on a fractional body weight basis (VC) for X compartment) were scaled with polynomial equations of various orders (n) as a function of body weight (BWT [g]), where A_n is the coefficient for each expression of the polynomial (Eq. 2). This framework could be used to define and switch among compartments of various species and genders with ease using one standard model.

$$\text{VC}_x = A_n \times \text{BWT}^n + A_{n-1} \times \text{BWT}^{n-1} + \dots + A_2 \times \text{BWT}^2 + A_1 \times \text{BWT} + A_0 \quad (2)$$

Oral absorption of chlorpyrifos is described by a two-compartment empirical model representing a stomach linked to an external intestinal compartment, which is linked to the liver compartment. Movement of chlorpyrifos among these absorption compartments are all described as first-order transfer rates including from stomach (Stom) to intestine (Int) (k_{si}) and intestine to liver (k_{ai}) (Eqs. (3) and (4)). This two-compartment is required to accurately describe chlorpyrifos absorption kinetics (Timchalk et al., 2002).

$$\frac{dA_{\text{Stom}}}{dt} = -k_{si} \times A_{\text{Stom}} \quad (3)$$

$$\frac{dA_{\text{Int}}}{dt} = k_{si} \times A_{\text{Stom}} - k_{ai} \times A_{\text{Int}} \quad (4)$$

Once absorbed, chlorpyrifos and chlorpyrifos-oxon were distributed among tissue compartments mediated by blood circulation described as a flow-limited process using previously published mathematical descriptions (Timchalk et al., 2002). The fraction of the compound not bound to plasma proteins (simply described mathematically as a fraction, Timchalk et al., 2002) was free to partition to various compartments described by partition coefficients.

Metabolism of chlorpyrifos and chlorpyrifos-oxon was implemented using *in vitro* to *in vivo* approach, in which metabolism, measured *in vitro*, was extrapolated to *in vivo* values using microsomal concentration and compartmental volume to scale metabolism (Lipscomb and Poet, 2008). Chlorpyrifos metabolism by cytochrome P450-mediated processes in the intestine, liver, and

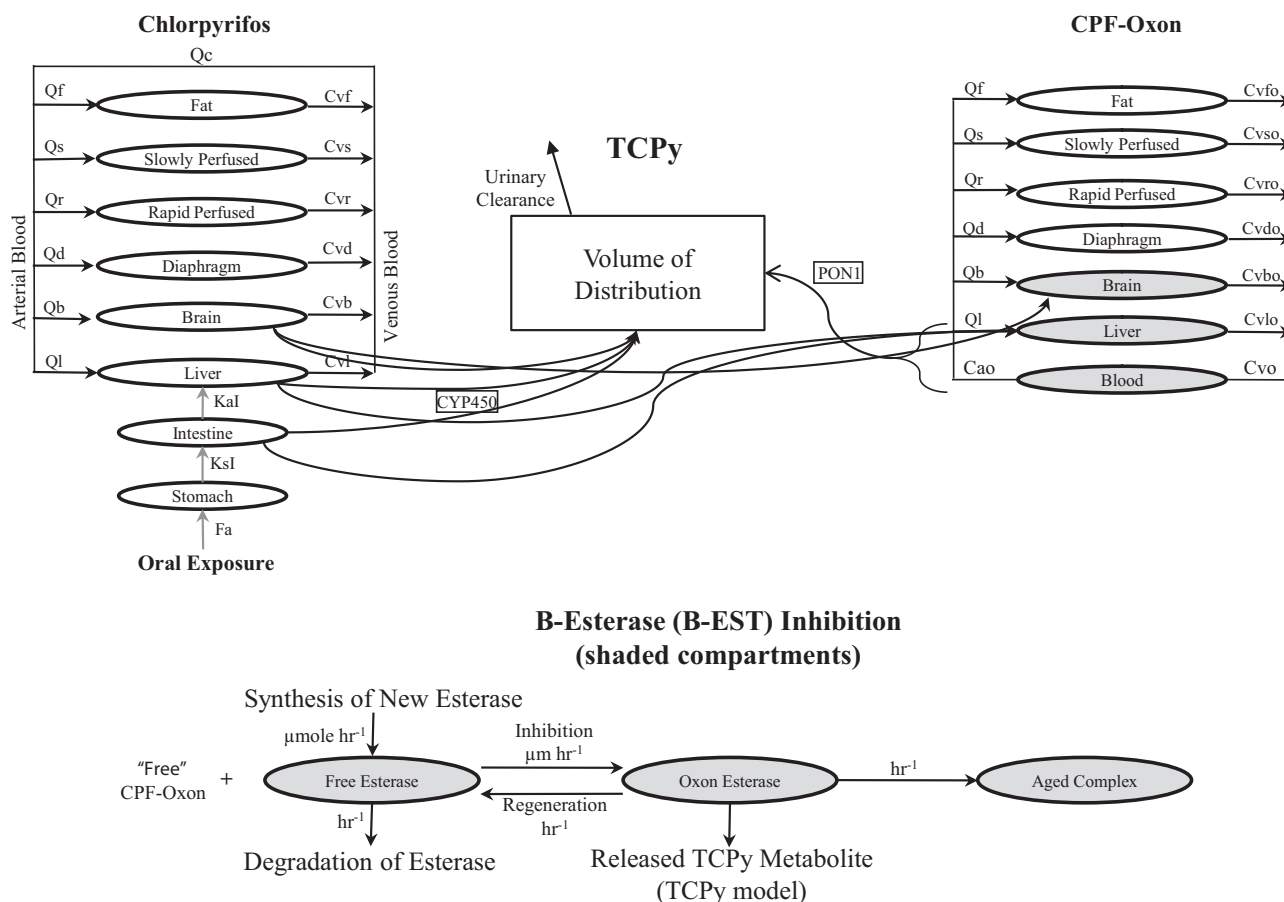


Fig. 2. The life-stage physiologically based pharmacokinetic and pharmacodynamic (PBPK/PD) model structure used to describe the disposition of chlorpyrifos (CPF), chlorpyrifos-oxon (CPF-oxon), and 3,5,6-trichloro-2-pyridinol (TCPy). The model also describes inhibition of B-esterases by chlorpyrifos-oxon in compartments shaded gray.

brain were described using Michaelis–Menten kinetics. Chlorpyrifos-oxon metabolism by PON1 was also described using Michaelis–Menten kinetics in those same compartments as chlorpyrifos (except brain) as well as in plasma.

B-esterase metabolism and inhibition were described in the liver, plasma, red blood cells (RBC), brain, and diaphragm as second-order processes. Tissue B-esterase enzyme levels and binding sites were calculated from enzyme turnover rates and enzyme activity levels and were balanced based on degradation, synthesis, and inhibition rates. Inhibition of B-esterase was calculated as described in previous models (Timchalk et al., 2002) using stoichiometric inhibition by chlorpyrifos-oxon concentration (based on an apparent bimolecular inhibition constant), reactivation of inhibited enzyme, aging of inhibited enzyme, and degradation/synthesis of new enzyme.

In order to describe changes in anatomy during growth, the life-stage PBPK/PD model dynamically changes during a simulation of the individual over its life span changing body weight, compartmental volumes, and all values scaled to those components. To accommodate the new description of dynamic compartment volumes, the mathematical description of percent inhibition also had to be altered from the original description (Timchalk et al., 2002). In the original chlorpyrifos PBPK/PD model, amount of uninhibited enzyme was divided by the total number of binding sites to calculate percent esterase activity (Timchalk et al., 2002). In the life-stage model, a naïve level of uninhibited enzyme is calculated based on the total number of binding sites changing over time by only degradation and synthesis of new enzyme (K_d ; Eqs. (5) and (6)). The ratio of the uninhibited enzyme, as calculated by Timchalk et al. (2002), to these naïve levels result in the calculation

of percent esterase activity, which remains at a constant 100% in a naïve (i.e. unexposed) simulation.

$$\frac{dAChE_{Naive}}{dt} = K_s - AChE_{Naive} \times K_d \quad (5)$$

$$\% \text{Activity} = \frac{AChE}{AChE_{Naive}} \times 100 \quad (6)$$

TCPy pharmacokinetics was described as a one-compartment model (Timchalk et al., 2002). All available data for human TCPy levels in urine were measured following acid hydrolysis of TCPy conjugates (Kisicki et al., 1999; Nolan et al., 1984; Timchalk et al., 2002), thus chlorpyrifos elimination was assumed to be completely facilitated as urinary TCPy.

2.4. Model parameters

Model parameters consisted of physiological values, partition coefficients, and biochemical values that were previously published, experimentally derived, or optimized computationally. Parameters of the Gompertz body weight function (Table 1) and polynomial descriptions of compartment volumes (Table 2) were obtained from previously published sources (Luecke et al., 2007; Young et al., 2009). The model developed here (which is designed for a “typical” person) is being used in an allied study to assess implications of model uncertainty and inter-individual variation in a risk assessment context (Poet et al., 2013). There is a significant portion of our population that have body weights higher than what could be used to reliably used to predict organ volumes with previous methods. Thus an effort was made to extend organ

Table 1

Parameter values for a Gompertz growth function used to describe body weight as a function of age in the life-stage model (Luecke et al., 2007).

Parameter	Female	Male
t_0	0	0
t_1	2.9490	2.9490
t_2	11	11
t_3	21.8653	21.6602
β_0, β_2	0	0
β_1	2.6727	2.4624
β_3	0.3113	-0.0911
W_0, W_1	3.4	3.4283
W_2, W_3	34.8806	34.3545
R_0, R_1	2.0740	2.1882
R_2, R_3	0.2128	0.1798
α_0, α_1	1.5046	1.5063
α_2, α_3	0.4438	0.2212

volume polynomial fits to larger body weights (~170 kg), including fat (both male and female), adipose, brain, liver, and muscle. Previously published polynomials or constants were used to define all other compartmental volumes (Table 2). The polynomial fitting procedure used here for refits was similar to that used previously with some minor modifications (Luecke et al., 2007; Young et al., 2009). Polynomial models of orders 1–6 were fit to available data, and then the best fit model was chosen based on the model with the lowest Bayesian information criterion (BIC). For data, emphasis was placed on reference values for various compartments as defined by the ICRP Publication 89 (2002), which presents detailed information on age- and gender-related differences in anatomical and physiological characteristics of reference individuals. This emphasis was placed especially for body weights from birth to ~70 kg, while other references were used for compartment volume data at larger body weights. Besides the ICRP Publication 89 (2002), Luecke et al. (2007) and Young et al. (2009), two other references were utilized for compartmental refits. First, a study aimed at evaluating differences in body mass and composition in respect to muscle strength and output between obese and normal weighted individuals, was used for fat and adipose tissue refits (Lafortuna et al., 2005). A study measuring skeletal muscle mass and distribution men and women with magnetic resonance imaging was used for muscle refits (Janssen et al., 2000). Data for some compartments were not available up to the desired ~170 kg. For those compartments, extrapolations were made with published relationships extrapolating data points for body weights of 150–175 kg at 5 kg intervals.

Other parameters determining pharmacokinetics included transfer rates and partition coefficients (Table 3). Fraction of dose

absorbed was determined from a non-compartmental pharmacokinetic analysis of human urinary excretion rate data based on the assumption that all chlorpyrifos is eliminated as TCPy in urine (Nolan et al., 1984). The total molar amount of TCPy eliminated was divided by the ratio of the area under the curve (AUC) from 0 to the time of the last data point to the AUC from 0 to ∞ resulting in the amount absorbed. The elimination constant for TCPy (k_e) was calculated from a non-compartmental pharmacokinetic analysis of TCPy levels in human blood (Nolan et al., 1984). The volume of distribution for TCPy was calculated utilizing partitioning coefficients predicted using an algorithm (Schmitt, 2008). This approach calculates partitioning coefficients based upon the composition of tissues in terms of water, various types of lipid content, phospholipid content, proteins, and pH differences using chemical compound specific lipophilicity, pKa, and plasma protein-binding (Schmitt, 2008; Smith et al., 2010). Tissue-specific partitioning coefficients were scaled to the appropriate compartment volume over various body weights. The volume of distribution was calculated as the total amount of TCPy in all compartments divided by the TCPy concentration in blood. To ease coding of this volume of distribution in the model, an allometric equation was fit to the predicted volume of distributions over various body weights (Eq. 7), where A and B are allometric constants.

$$VD_{\text{TCPy}} = A \times \text{BWT}^B \quad (7)$$

The transfer rate of chlorpyrifos from stomach to intestine, absorption rate from intestine to liver of chlorpyrifos, were optimized in tandem using TCPy levels in blood and plasma cholinesterase inhibition (Nolan et al., 1984). Due to differences in dose formulations, model simulations of the two human studies used different absorption parameters. For the Kisicki study, the stomach to intestine transfer rate was optimized independently using levels of TCPy in blood (Kisicki et al., 1999; Timchalk et al., 2002). Transfer rate of chlorpyrifos-oxon and TCPy from intestine to liver was optimized in a rat model using rats dosed with TCPy (Busby-Hjerpe et al., 2010; Marty et al., 2012; Timchalk et al., 2007a; Timchalk et al., 2002). Blood flows to each compartment were scaled as a constant flow per compartmental volume, and total cardiac output was a summation of all blood flows to all compartments. Fractional binding of chlorpyrifos to serum albumin was measured previously using a dialysis system (Lowe et al., 2009). As chlorpyrifos-oxon undergoes rapid hydrolysis in the presence of albumin, it was assumed that chlorpyrifos and chlorpyrifos-oxon have equivalent fraction binding in plasma. Partition coefficients were measured previously using plasma and vegetable oil and extrapolated to tissues using a QSAR approach (Lowe et al., 2009).

Table 2

Parameter values for polynomial functions describing compartment volume (fraction of body weight) as a function of body weight (g) used in the life-stage model.

Compartment	A_0	A_1	A_2	A_3	A_4	A_5	A_6	Source
Brain	1.216×10^{-1}	-3.465×10^{-6}	4.354×10^{-11}	-2.463×10^{-16}	5.132×10^{-22}			Fit (see text)
Hepatic	3.939×10^{-2}	-7.058×10^{-7}	1.155×10^{-11}	-8.016×10^{-17}	1.869×10^{-22}			Fit (see text)
Blood	8.970×10^{-2}	-3.500×10^{-7}	6.540×10^{-13}					Young et al. (2009)
Fat-male	3.484×10^{-2}	2.803×10^{-5}	-1.422×10^{-9}	2.892×10^{-14}	-2.718×10^{-19}	1.203×10^{-24}	-2.036×10^{-30}	Fit (see text)
Fat-female	9.217×10^{-2}	1.401×10^{-5}	-6.787×10^{-10}	1.540×10^{-14}	-1.558×10^{-19}	7.249×10^{-25}	-1.274×10^{-30}	Fit (see text)
Adipose	2.044×10^{-1}	2.617×10^{-5}	-1.542×10^{-9}	3.268×10^{-14}	-3.116×10^{-19}	1.387×10^{-24}	-2.35×10^{-30}	Fit (see text)
Lung	1.860×10^{-2}	-4.550×10^{-8}						Young et al. (2009)
Kidney	7.260×10^{-3}	-6.690×10^{-8}	3.330×10^{-13}					Young et al. (2009)
Spleen	3.120×10^{-3}	-5.570×10^{-9}						Young et al. (2009)
GI	1.650×10^{-2}							Brown et al. (1997)
Muscle	1.251×10^{-1}	1.458×10^{-5}	-2.927×10^{-10}	2.114×10^{-15}	-5.250×10^{-21}			Fit (see text)
Skin	1.030×10^{-1}	-2.560×10^{-6}	3.680×10^{-11}	-2.580×10^{-16}	8.620×10^{-22}	-1.100×10^{-27}		Young et al. (2009)
Bone marrow	2.100×10^{-2}							Brown et al. (1997)
Diaphragm	3.000×10^{-4}							Brown et al. (1997) and Luecke et al. (2007)
Pancreas	1.480×10^{-3}							Young et al. (2009) and Brown et al. (1997)

Table 3

Values for parameters describing the disposition of chlorpyrifos (CPF), chlorpyrifos-oxon (CPF-oxon), and 3,5,6-trichloro-2-pyridinol (TCPy) in the life-stage model.

Parameter	Value	Source
<i>Blood flow (L/h/kg tissue)</i>		
Brain	30.6	Price et al. (2003)
Liver	50.4	Price et al. (2003)
Fat	1.45	Luecke et al. (2007), Cowles et al. (1971) and Price et al. (2003)
Rapidly perfused	61.8	Adrenal/Spleen mean: Price et al. (2003)
Slowly perfused	1.8	Bone marrow: Price et al. (2003)
Diaphragm	85.2	Luecke et al. (2007)
<i>Partition coefficients (tissue:blood)</i>		
CPF		
Brain	16.5	Lowe et al. (2009)
Liver	12.8	Lowe et al. (2009)
Fat	250	Lowe et al. (2009)
Rapidly perfused	16.5	Lowe et al. (2009)
Slowly perfused	3.85	Lowe et al. (2009)
Diaphragm	3.85	Lowe et al. (2009)
CPF-oxon		
Brain	5.6	Lowe et al. (2009)
Liver	4.5	Lowe et al. (2009)
Fat	75	Lowe et al. (2009)
Rapidly perfused	5.6	Lowe et al. (2009)
Slowly perfused	1.8	Lowe et al. (2009)
Diaphragm	1.8	Lowe et al. (2009)
Hepatic dearylation		
$V_{\max}$ ($\mu\text{mol/h/kg tissue}$)	1360	Smith et al. (2010) and Wilson et al. (2003)
K_m (μM)	21	Smith et al. (2010), Poulin and Haddad (2011) and Poulin et al. (2011)
Hepatic desulfuration		
$V_{\max}$ ($\mu\text{mol/h/kg tissue}$)	630	Smith et al. (2010) and Wilson et al. (2003)
K_m (μM)	33.2	Smith et al. (2010), Poulin and Haddad (2011) and Poulin et al. (2011)
Hepatic CPF-oxon hydrolysis		
$V_{\max}$ ($\mu\text{mol/h/kg tissue}$)	1.29×10^5	Smith et al. (2010) and Wilson et al. (2003)
K_m (μM)	192	Smith et al. (2010), Poulin and Haddad (2011) and Poulin et al. (2011)
Blood CPF-oxon hydrolysis		
$V_{\max}$ ($\mu\text{mol/h/kg tissue}$)	Logistic Fit (see text)	Smith et al. (2010)
K_m (μM)	192	Smith et al. (2010)
Intestinal dearylation		
$V_{\max}$ ($\mu\text{mol/h/kg tissue}$)	53	Poet et al. (2003)
K_m (μM)	21	Measured in liver Smith et al. (2010), Poulin and Haddad (2011) and Poulin et al. (2011)
Intestinal CPF-oxon hydrolysis		
$V_{\max}$ ($\mu\text{mol/h/kg tissue}$)	82	Poet et al. (2003)
K_m (μM)	192	Measured in liver Smith et al. (2010), Poulin and Haddad (2011) and Poulin et al. (2011)
Intestinal desulfuration		
$V_{\max}$ ($\mu\text{mol/h/kg tissue}$)	28	Poet et al. (2003)
K_m (μM)	33.2	Measured in liver Smith et al. (2010), Poulin and Haddad (2011) and Poulin et al. (2011)
Brain dearylation		
$V_{\max}$ ($\mu\text{mol/h/kg tissue}$)	3.85	Extrapolated (see text)
K_m (μM)	21	Measured in liver Smith et al. (2010), Poulin and Haddad (2011) and Poulin et al. (2011)
Brain desulfuration		
$V_{\max}$ ($\mu\text{mol/h/kg tissue}$)	0.91	Extrapolated (see text)
K_m (μM)	33.2	Measured in liver Smith et al. (2010)
Transfer rates (/h)		
CPF stomach to intestine	0.31/0.086	Fit to Nolan et al. (1984)/Kisicki et al. (1999)
Intestinal absorption to liver		
CPF	3.1	Fit to Nolan et al. (1984)
CPF-oxon	3.9	Fit to rat data Marty et al. (2012)
TCPy	8.5	Fit to rat data Timchalk et al. (2007a) and Busby-Hjerpe et al. (2010)
Plasma protein binding (%)		
CPF	99	Lowe et al. (2009)
Oxon	99	Lowe et al. (2009)
TCPy compartmental model		
K_e (/h)	0.024	Nolan et al. (1984)
Volume of distribution constant A	0.176	Fit to Schmitt et al. (2008)
Volume of distribution constant B	1.01	Fit to Schmitt et al. (2008)

Metabolic parameters were fit or acquired from several sources (Table 3). Hepatic chlorpyrifos and chlorpyrifos-oxon metabolism were measured previously in human microsomes from donors of a wide age range (13 d–75 yr) and demonstrated a consistent level of metabolism across ages (Smith et al., 2011). Chlorpyrifos-oxon metabolism in plasma was also measured in humans (3 d–43 yr), and $V_{\max}$ values demonstrated an age-dependent increase (Smith et al., 2011). Thus $V_{\max}$ was scaled using a logistic model as a function of age (Eq. 8), where A, B, and C are parameters. Hepatic *in vitro*

$V_{\max}$ values were scaled using a constant microsomal protein level across all ages (33 mg/g tissue) (Wilson et al., 2003), and *in vitro* K_m values were scaled based upon predictions of unbound substrate in the *in vitro* system to that of the *in vivo* system (39% of measured K_m) (Poulin and Haddad 2011; Poulin et al., 2011). In order to determine the relative contribution of chlorpyrifos-oxon hydrolysis by A- and B-esterases, samples of human plasma (200 μL) were spiked with chlorpyrifos-oxon ranging from 0.1 nM to 10 μM , including a control (0 nM). After 120 min, cholinesterase activity

was measured using the Ellman assay (Ellman et al., 1961). Intestinal metabolism of chlorpyrifos and chlorpyrifos-oxon was measured in rat enterocytes (Poet et al., 2003), and $V_{\max}$ values were extrapolated to humans. K_m values were assumed to be equivalent to the K_m values measured in liver from humans (Smith et al., 2011). Chlorpyrifos metabolism in brain was estimated based on an extrapolation from limited studies *in vitro* with rat tissue (Timchalk et al., 2012). Chlorpyrifos dearylation in brain microsomes was 115-fold lower than in hepatic microsomes. Formation of chlorpyrifos-oxon in brain microsomes was below the limit of detection (30 ng/mL). Thus chlorpyrifos metabolism in brain was estimated at 115 times lower than hepatic chlorpyrifos metabolism (for both desulfuration and dearylation) on a microsomal protein basis.

$$V_{\max} = \frac{A}{1 + e^{\frac{B - \text{age}}{c}}} \quad (8)$$

Pharmacodynamic parameters were acquired from a variety of literature sources and included cholinesterase activities, enzyme aging, degradation, reactivation, and turnover rates (Table 4) (Albers et al., 2010; Hojring and Svensmark, 1976; Li et al., 2005; Maxwell et al., 1987; Mortensen et al., 1998b; Pope et al., 2005; Sidell and Kaminskis, 1975). Verification of the bimolecular inhibition rate constants was achieved by directly determining the acetylcholinesterase and butyrylcholinesterase inhibition dynamics in rat plasma, brain, and saliva (Kousba et al., 2007; Kousba et al., 2004).

2.5. Sensitivity analysis

A local sensitivity analysis was conducted to identify the most important parameters for estimating acetylcholinesterase inhibition in red blood cells and brain from an oral dose of 3 µg/kg chlorpyrifos (the reference dose) in 0.5 and 30 yr humans. In order to conduct the analysis, age-dependent parameters defined as equations were replaced with discrete constants. All variables in the model were subjected to the sensitivity analysis. Normalized sensitivity coefficients were calculated for a 1% change in a given model parameter when all other parameters were held fixed. A more rigorous sensitivity analysis will be implemented in future work to gauge the implications of parameter uncertainty and inter-variability (Poet et al., 2013).

2.6. Statistics and model coding

The PBPK/PD model was coded in acslXtreme 3.0.2.1 (Aegis Technology, Huntsville, AL USA) and included ordinary differential equations and algebraic functions. Optimizations of PBPK/PD model parameters were also conducted using acslXtreme using the Nelder-Mead or Quasi-Newton (if less than two variables being optimized) algorithms. The heteroscedasticity was set to 1.0, except for TCPy levels in blood, which was set to 0.5, because initial fits with a heteroscedasticity factor of 1.0 did not fit early time points well visually. Initial values were set by adjusting parameters visually, then allowing the optimization to occur bounded by reasonable values. R: A language and environment for statistical computing, version 2.9.0, 2.11.1, or 2.13.1 (R Foundation for Statistical

Table 4
Values for parameters describing B-esterase activity and inhibition thereof by chlorpyrifos-oxon (CPF-oxon) in the life-stage model.

Parameter	Value	Source
Enzyme activity (µmol/kg/h)		
AChE		
Brain	4.4×10^5	Maxwell et al. (1987)
Plasma	NA	Timchalk et al. (2002)
Liver	1.02×10^4	Maxwell et al. (1987)
Red blood cells	4.27×10^5	Albers et al. (2009)
Diaphragm	7.74×10^4	Maxwell et al. (1987)
BuChE		
Brain	4.68×10^4	Maxwell et al. (1987)
Plasma	2.63×10^5	Sidell and Kaminskis (1975)
Liver	3.0×10^4	Maxwell et al. (1987)
Diaphragm	2.64×10^4	Maxwell et al. (1987)
Carboxylesterase		
Brain	2.88×10^5	Hojring et al. (1976)
Plasma	NA	Li et al. (2005)
Liver	1.27×10^6	Pope et al. (2005)
Diaphragm	3.18×10^5	Maxwell et al. (1987)
Degradation rates (/h)		
AChE	0.01	Timchalk et al. (2002) and Gearhart et al. (1990)
BuChE	0.0024	Fit to Nolan et al. (1984)
Carboxylesterase	0.001	Timchalk et al. (2002); Fit to Chanda et al. (1997)
Brain	7.54×10^{-4}	Timchalk et al. (2002); Fit to Chanda et al. (1997)
Enzyme reactivation rate (/h)		
Acetyl	0.014	Carr and Chambers (1996) and Timchalk et al. (2002)
Butyryl	0.0014	Carr and Chambers (1996); Timchalk et al. (2002)
Carboxyl	0.014	Carr and Chambers (1996); Timchalk et al. (2002)
Enzyme Aging Rate (/h)	0.0113	Carr and Chambers (1996); Timchalk et al. (2002)
Enzyme turnover rate (/h)		
Acetyl	1.17×10^7	Maxwell et al. (1987)
Butyryl	3.66×10^6	Maxwell et al. (1987)
Carboxyl	1.086×10^5	Maxwell et al. (1987)
Bimolecular inhibition rate (µM/h)		
Acetyl	220	Kousba et al. (2004) and Kousba et al. (2007)
Red blood cells	100	Timchalk et al. (2002)
Butyryl	2000	Timchalk et al. (2002)
Carboxylesterase	20	Timchalk et al. (2002); Fit to Chanda et al. (1997)

Computing, Vienna, Austria) was used to analyze data, calculate AUC values, and fit various functions including polynomial descriptions of compartments, logistic function for chlorpyrifos-oxon hydrolysis $V_{\max}$, and two-compartment pharmacokinetic models. For summary statistics, geometric means are reported with 90% confidence intervals in parentheses.

3. Results

3.1. Parameterization-compartment volumes

The parameterized Gompertz function for body weights of males and females (Luecke et al., 2007) provided good predictions of 2007–2008 US population body weight from the National Health and Nutrition Examination Survey (NHANES) (Supplemental Fig. 1).

Most literature compartment fits accurately described age-dependent growth for body weights up to ~170 kg; however due to the unpredictable nature of polynomials outside of the range of fitted data, fat (both male and female), adipose, brain, liver, and muscle compartments were all refitted for extrapolations of body weights up to ~170 kg (Fig. 3, Table 2). Raw data and extrapolations of data (150–175 kg body weights) were used to fit male and female fat compartments (Supplemental Figs. 2A and B) (2002; Lafortuna et al., 2005). Data from these same sources were used for adipose tissue, based on the assumption that 80% of adipose tissue in adults is fat, as did ICRP Publication 89 in their derivation of reference adipose levels for adults (2002). Resulting fits were 6th order polynomials for male fat and adipose compartments and a 6th order polynomial was fit for the female fat compartment (even though it did not have the lowest BIC) to maintain consistency with males. Source data for the brain compartment was acquired for body weights less than (2002) and greater than 70 kg (Young et al., 2009). Extrapolations of brain volume were made for using 150–175 kg body weights using a table provided by Young et al., (2009). The resulting fit was a 4th order polynomial equation (Supplemental Fig. 3A). The liver compartment was refitted using reference values for liver volumes as well as extrapolated for 150–175 kg body weights as previously for brain volume (Supplemental Fig. 3B) (2002; Young et al., 2009). Originally, all data were used in fitting attempts (Young et al., 2009), but those efforts were problematic. Those fits resulted in polynomial equations that attempted to explain variability in liver weights, and thus a smooth function across body weights was

unattainable. Reference values for liver volumes resulted in a 4th order polynomial that was a much smoother fit. Raw muscle data (2002; Janssen et al., 2000) and extrapolations of body weights 150–170 kg based on the equation presented in Janssen et al. (2000) were used to fit a 4th order equation that is summed into the slow compartment (Supplemental Fig. 3C). These compartment fits in addition to the other compartments from various sources provide a total volume of distribution that is ~75–95% of body weight depending on age (Supplemental Fig. 4).

The TCPy volume of distribution was predicted over various body weights using a partitioning algorithm (Schmitt, 2008). Partitioning of TCPy to each compartment in the PBPK model was predicted and the amount of TCPy in each compartment was calculated using compartmental volumes for various body weights 3–170 kg calculated using polynomial scaling as previously described. For each body weight, the ratio of total TCPy in all compartments to the TCPy concentration in blood was used to calculate the TCPy volume of distribution (Supplemental Fig. 5). An allometric equation was fit to these predictions for easier coding into the PBPK model (Eq. 7, A: 0.176, B: 1.01) (Table 3).

3.2. Parameterization-other parameters

Since blood flows were scaled as constant flow per compartment volume, they demonstrated similar ontogeny patterns as compartment volumes (Fig. 4). In 30 yr old adults, highest overall blood flows were to the rapidly perfused compartment; while in neonates, the highest blood flow was to the brain.

Previously measured chlorpyrifos-oxon hydrolysis $V_{\max}$ values demonstrated an age-dependent increase in plasma (Smith et al., 2011), and a logistic model (Eq. 8) was fit to the data with parameter values of 4.32×10^5 , 3.052, and 3.90 for the logistic parameters A, B, and C, respectively (Supplemental Fig. 6). This mathematical description describes age-dependent description of $V_{\max}$ values over the ontogeny of this enzymatic system. Since both A- and B-esterases are present in plasma, cholinesterase inhibition was measured in undiluted human plasma after incubation with chlorpyrifos-oxon to determine the relative contribution of each esterase to chlorpyrifos-oxon hydrolysis. After 2 h, 10 nM of chlorpyrifos-oxon inhibited plasma cholinesterase activity by ~10%, and nearly all cholinesterase activity was inhibited with 500 nM chlorpyrifos-oxon (Supplemental Fig. 7). This indicates that the relative contribution of chlorpyrifos-oxon hydrolysis by B-esterases was negligible compared to A-esterases at concentrations used to

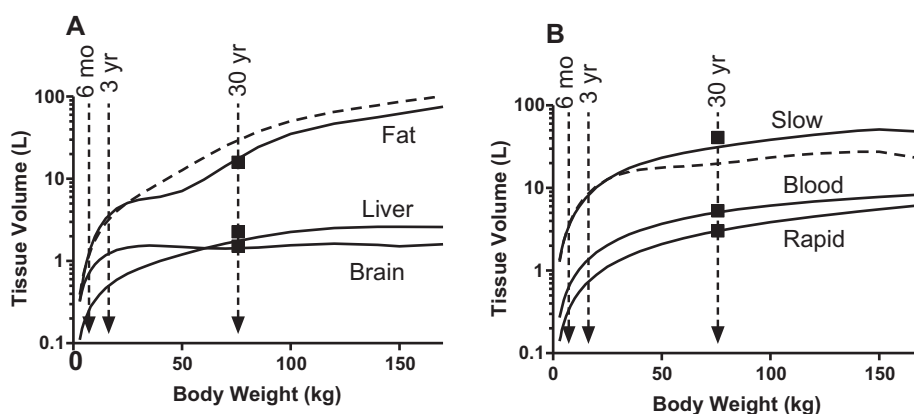


Fig. 3. Compartment volumes as a function of body weight for males/gender neutral (solid lines) and females (dotted lines) of the life-stage model as calculated by polynomial equations of fractional body weight (Luecke et al., 2007; Young et al., 2009). Symbols represent tissue volumes predicted for a typical 70 kg male using the original chlorpyrifos PBPK/PD model (Brown et al., 1997; Timchalk et al., 2002). Arrows indicate body weights of typical humans at 6 months, 3 yr, and 30 yr old as determined by a generalized Gompertz equation (Luecke et al., 2007). Note: y-axes are on a logarithmic scale.

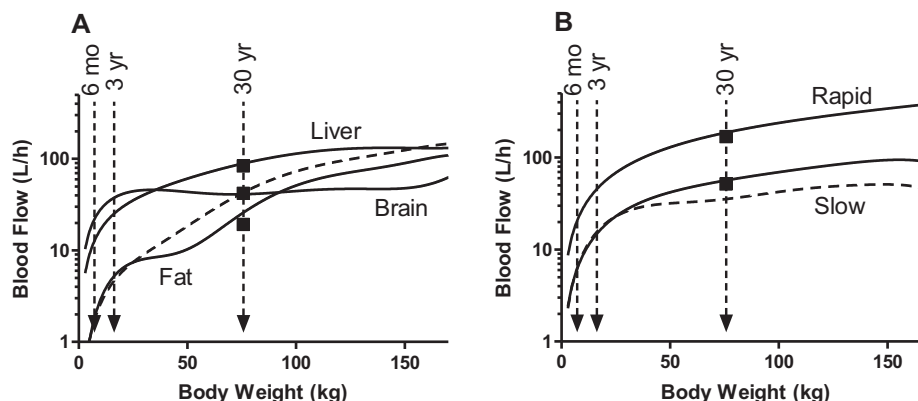


Fig. 4. Blood flow to various compartments as a function of body weight for males or both genders (solid lines) and females (dotted lines) of the life-stage model. Symbols represent blood flows for a typical 70 kg male using standard allometric approaches (Brown et al., 1997; Timchalk et al., 2002). Arrows indicate body weights of humans at 6 months, 3 yr, and 30 yr old as determined by a generalized Gompertz equation (Luecke et al., 2007). Note: y-axes are on a logarithmic scale.

measure the *in vitro* Michaelis–Menten metabolism kinetics in diluted (1/200) human plasma (72.3–2400 μM) (Smith et al., 2011).

Fraction of chlorpyrifos dose absorbed was determined in humans using ratios of AUC values [$\text{AUC}_{(0 \text{ to } t_{\text{last}})}$ to $\text{AUC}_{(0 \text{ to } \infty)}$] of urinary excretion rates and total amount of TCPy eliminated. After administration to chlorpyrifos on a lactose tablet, subjects demonstrated a fractional absorption of 77% (Nolan et al., 1984), while those that were administered neat chlorpyrifos in a capsule absorbed 21% of the dose (Kisicki et al., 1999; Timchalk et al., 2002).

Pharmacokinetic and cholinesterase inhibition data from Nolan et al. (1984) was used to calibrate the PBPK/PD model. TCPy was measured in all blood and urine samples. Chlorpyrifos in blood was also measured very close to the limits of detection and displayed erratic temporal behavior. The chlorpyrifos transfer rate from stomach to intestine (k_{si} , 0.31 h^{-1}) and the chlorpyrifos absorption transfer rate from intestine to liver (k_{ai} , 3.1 h^{-1}) was fit using TCPy levels in blood and plasma cholinesterase inhibition data. Since chlorpyrifos can be metabolized in the GI tract, ratios of these parameters can alter the relative absorption of chlorpyrifos and TCPy, with blood levels of chlorpyrifos being relatively sensitive to k_{si} compared to k_{ai} . The degradation rate of butyrylcholinesterase was fit to the plasma cholinesterase inhibition data (Nolan et al., 1984).

Ultimately, simulations of oral chlorpyrifos doses resulted in levels of TCPy in blood and urine that were in agreement with previous observed values. The peak geometric mean TCPy concentration (with 90% confidence interval) in blood was 4.7 (3.4–6.4) μM at 6 h post ingestion (Nolan et al., 1984). At the same dose and

time, the life-stage PBPK/PD model predicts a TCPy concentration of 4.5 μM in blood of a 30 year old male and a slightly higher peak level at 9.4 h (Fig. 5A). In urine, 70% of the total dose (83.3 (70.5–98.6) μmol) was excreted by 120 h (Nolan et al., 1984), and the model predicts 72% (85.5 μmol) for the same metric (Fig. 5B).

Life-stage PBPK/PD predictions of cholinesterase inhibition are very similar to those measured previously in adult humans. Nolan et al. (1984) measured peak inhibition of plasma cholinesterase activity (geometric mean: 17 [13–21 90% confidence intervals]% control activity) at 24 h and no significant changes to red blood cell cholinesterase following an oral dose of 0.5 mg/kg chlorpyrifos. Substantial recovery of plasma cholinesterase (82 [68–99]% control cholinesterase activity) occurred by 648 h. The life-stage PBPK/PD model predicts 19% control plasma cholinesterase activity at 24 h, and peak inhibition (17% control activity) at 13.7 h (Fig. 8A). The model also predicts 92% of control cholinesterase activity in red blood cells at peak inhibition (11.5 h, data not shown). The life-stage model predicted cholinesterase activity to recover at a similar rate as observed data, as the model predicts 84% control plasma cholinesterase activity at 648 h.

3.3. Validation in adult humans

A different study was used to validate the model for adults, in which adult human volunteers were administered 0.5, 1, or 2 mg/kg chlorpyrifos in a capsule (Kisicki et al., 1999; Timchalk et al., 2002). Due to a different formulation of chlorpyrifos from the Nolan et al. (1984) study, the chlorpyrifos transfer rate from stomach to small intestine was fit to TCPy levels in blood of the 1 mg/kg dose group (Kisicki et al., 1999; Timchalk et al., 2002).

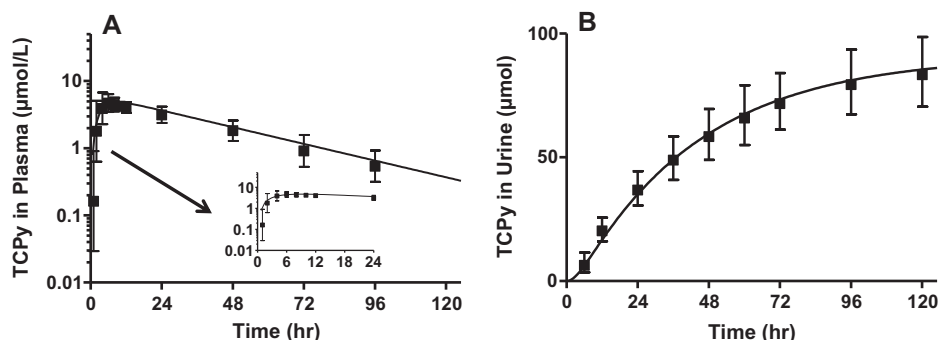


Fig. 5. Life-stage model predictions of 3,5,6-trichloro-2-pyridinol (TCPy) concentrations in plasma (A) and urine (B) from a human volunteer study (Nolan et al., 1984). Data represent geometric means with 90% confidence intervals for 5 individuals following an oral chlorpyrifos dose of 0.5 mg/kg (Nolan et al., 1984). Note: y-axis in (A) is on a logarithmic scale.

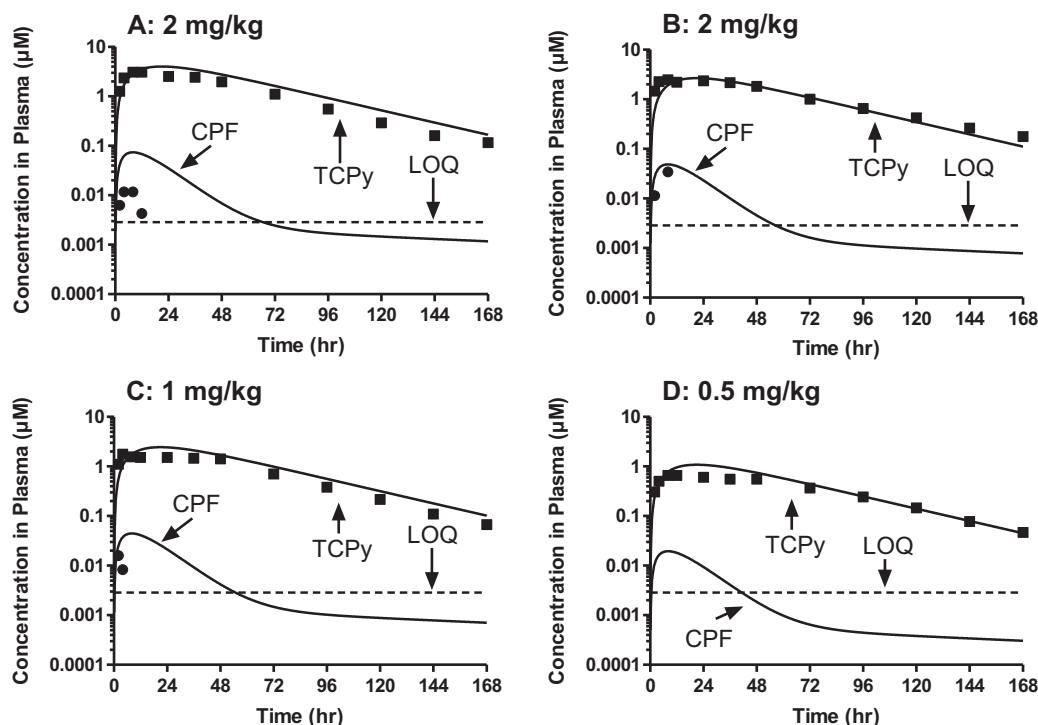


Fig. 6. Life-stage model predictions of chlorpyrifos (CPF) and 3,5,6-trichloro-2-pyridinol (TCPy) concentrations in four individuals from a human volunteer study (Kisicki et al., 1999; Timchalk et al., 2002). Individuals A and B were dosed with 2 mg/kg CPF (0.21 and 0.14 fractional absorption, respectively), individual C was dosed with 1 mg/kg CPF (0.26 fractional absorption), and individual D was dosed with 0.5 mg/kg CPF (0.23 fractional absorption). The limit of quantitation (LOQ) is indicated as a dotted line. Note: y-axis is on a logarithmic scale.

The life-stage PBPK/PD model was used to simulate chlorpyrifos and TCPy levels in plasma for several individuals using estimated absorbed doses determined from total TCPy excreted in urine. Chlorpyrifos levels were very near the limits of quantitation (0.003 μM), and several individuals in the 1 and 2 mg/kg dose group had quantifiable chlorpyrifos levels in blood. Using individualized fractions of absorption based on urinary TCPy elimination, the model slightly over predicted chlorpyrifos levels in blood (Fig. 6).

For summarized data sets, geometric mean (with 90% confidence intervals) peak TCPy concentrations in blood (8 h) were 0.8 (0.6–1.0), 1.5 (1.2–2.0), and 2.6 (1.9–3.2) μM , respectively for three dose groups (Kisicki et al., 1999; Timchalk et al., 2002). Using a fractional absorption of 0.21, the life-stage model predicts 0.7, 1.4, and 2.8 μM TCPy at that same time point (Fig. 7A), and urinary excretion is similar to those predicted by the model (Fig. 7B). Red blood cell cholinesterase activity displayed no significant

inhibition (Kisicki et al., 1999; Timchalk et al., 2002). The life-stage PBPK/PD model predicts 94% control cholinesterase activity at peak inhibition (27 h), which is similar to what was measured (Fig. 8B–D).

3.4. Life-stage predictions

The life-stage PBPK/PD model predicts age-dependent responses in pharmacokinetics and resulting cholinesterase inhibition in humans from chlorpyrifos exposure. PBPK/PD model simulations of 6 months, 3 yr, and 30 yr old humans at various doses (100% oral absorption) predict that adults will have the highest peak chlorpyrifos levels in blood (Figs. 9A and 10A). At single high oral doses ≥ 0.6 mg/kg, 6 months infants will have the highest peak levels of chlorpyrifos-oxon in blood (e.g. $\sim 55\%$ higher at 1 mg/kg chlorpyrifos), and below that, adults will have the highest levels (Figs. 9B and 10B). Higher levels of chlorpyrifos-oxon result

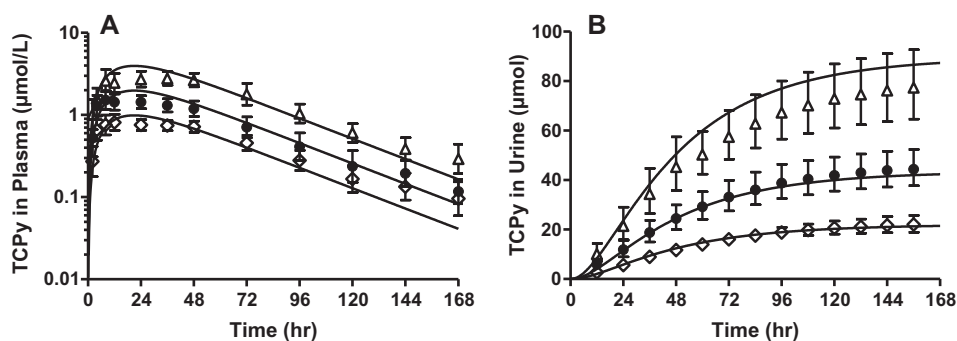


Fig. 7. Life-stage model predictions of 3,5,6-trichloro-2-pyridinol (TCPy) concentrations in plasma (A) and urine (B) from a human volunteer study (Kisicki et al., 1999; Timchalk et al., 2002). Data represent geometric means with 90% confidence intervals for 12 individuals dosed with 0.5 (diamond), 1.0 (circle), or 2 mg/kg (triangle) CPF. Note: y-axis in 7B is on a logarithmic scale.

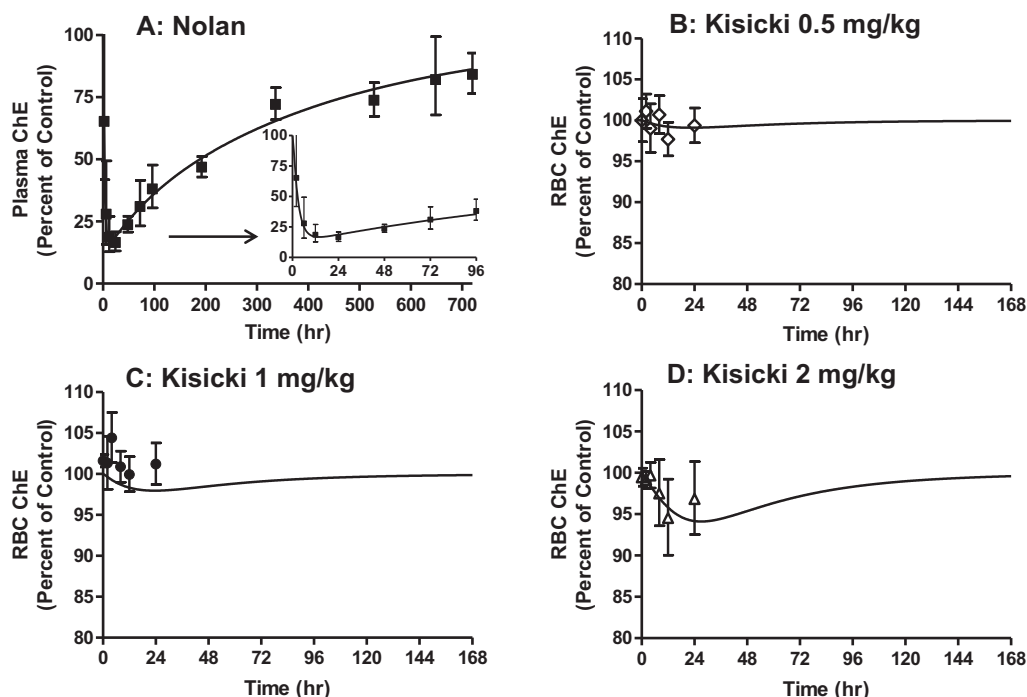


Fig. 8. Life-stage model predictions of cholinesterase (ChE) activities from human volunteer studies including plasma cholinesterase from a 0.5 mg/kg chlorpyrifos (CPF) (A) (Nolan et al., 1984) and red blood cell cholinesterase activity (Kisicki et al., 1999; Timchalk et al., 2002), after being dosed with 0.5 (B), 1.0 (C), or 2 mg/kg (D) CPF.

in higher amounts of peak red blood cell acetylcholinesterase (e.g. 13% more inhibition at 1 mg/kg chlorpyrifos) and plasma cholinesterase activity inhibition in infants compared to adults at doses ≥ 0.6 mg/kg chlorpyrifos (Figs. 9C, D and 10C). Plasma cholinesterase is inhibited at lower levels of chlorpyrifos-oxon than red blood cell acetylcholinesterase, and after an oral dose of 0.6 mg/kg chlorpyrifos, ~99% of plasma cholinesterase was predicted to be inhibited in both adults and 6 months infants (Fig. 9D). Thus the model predicts adults will have lower levels of plasma cholinesterase activity compared to infants at equivalent doses through the majority of the dynamic range of plasma cholinesterase inhibition.

In the brain, trends in peak chlorpyrifos concentrations are predicted to mirror those in blood, where adults have higher chlorpyrifos levels than 6 months olds (Fig. 11A). There are two sources of chlorpyrifos-oxon in the brain, chlorpyrifos-oxon formed from hepatic chlorpyrifos metabolism and chlorpyrifos-oxon formed locally due to low levels chlorpyrifos metabolism in the brain. Because of these two chlorpyrifos-oxon sources, higher levels of chlorpyrifos-oxon will be formed in the brains of adults compared to 6 months infants because of higher chlorpyrifos levels (with the assumption of no age-dependent differences in brain metabolism rates), and at doses ≥ 0.6 mg/kg chlorpyrifos, chlorpyrifos-oxon formed hepatically will be slightly greater in 6 months infants than adults. Overall, the model predicts slightly higher levels of chlorpyrifos-oxon in adults than in 6 months old infants (e.g. 8% higher at 5 mg/kg chlorpyrifos), suggesting that even low levels of localized chlorpyrifos metabolism may be important (Fig. 11B). Due to stoichiometric cholinesterase inhibition, peak inhibition levels of brain acetylcholinesterase activity follow the same pattern as that of peak chlorpyrifos-oxon levels. Since significant inhibition is not predicted to occur until ~5 mg/kg, the model predicts adults will have slightly increased brain acetylcholinesterase inhibition compared to children (Fig. 11C).

Simulations of time-course pharmacokinetics and biological response also result in age-dependent patterns. Following a single oral dose of 1 mg/kg chlorpyrifos, predicted time courses of

chlorpyrifos, chlorpyrifos-oxon, and inhibition of red blood cell cholinesterase activity follow similar patterns as the respective peak levels, where adults had the highest levels of chlorpyrifos (as for all doses), and 6 months infants had the highest levels of chlorpyrifos-oxon and red blood cell cholinesterase inhibition (for doses >0.6 mg/kg) (Fig. 12). Adults had a higher AUC value of chlorpyrifos concentration in blood and longer half-lives of chlorpyrifos in blood (Table 5). Infants of 6 months of age had a larger AUC value of chlorpyrifos-oxon in blood. AUC values of TCPy in blood and half-lives of chlorpyrifos-oxon and TCPy in blood were very similar.

3.5. Sensitivity analysis

A sensitivity analysis was conducted on the life-stage PBPK/PD model for peak inhibition of red blood cell and brain acetylcholinesterase activity. Of the selected parameters, red blood cell acetylcholinesterase inhibition for both ages was most sensitive to blood flow to liver, fraction of chlorpyrifos-oxon bound in plasma, chlorpyrifos-oxon partition coefficient in liver, acetylcholinesterase levels in red blood cells, and chlorpyrifos metabolism in liver (Table 6). Inhibition of acetylcholinesterase in brain for both ages was most sensitive to brain chlorpyrifos desulfuration, hepatic chlorpyrifos dearylation, hepatic chlorpyrifos-oxon hydrolysis, brain volume, blood flow to liver, acetylcholinesterase levels in the brain, carboxylesterase levels in brain, carboxylesterase levels in liver, fractions of both chlorpyrifos and chlorpyrifos-oxon bound in plasma, and chlorpyrifos partition coefficients in liver as well as brain (Table 6). Sensitive parameters were, for the most part, consistent between both age groups. The only major differences between age groups was for chlorpyrifos-oxon hydrolysis in blood and liver was sensitive in adults, where in 6 months infants, other sequestrations of chlorpyrifos-oxon were more sensitive (e.g. B-esterases). This could be due to higher capacities of chlorpyrifos-oxon hydrolysis in adults.

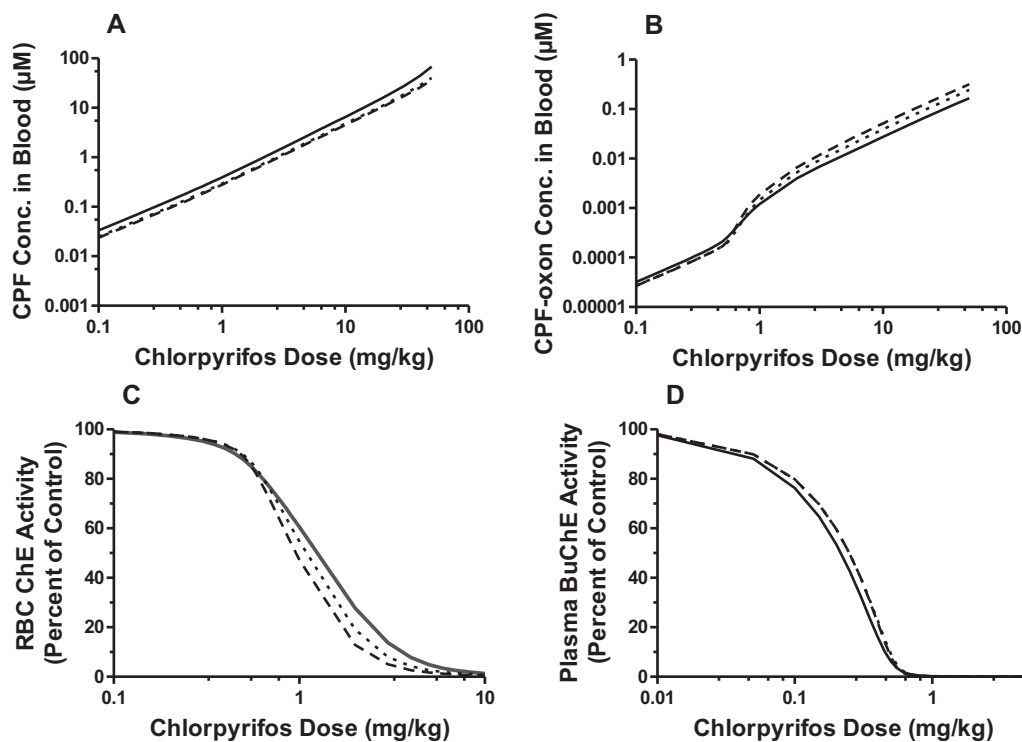


Fig. 9. Life-stage model predictions of 6 months infants (dashed line), 3 yr old children (dotted line), and 30 yr old adults (solid line) exposed to a variety of chlorpyrifos (CPF) doses. Predictions include peak chlorpyrifos concentration in blood (A), chlorpyrifos-oxon (CPF-oxon) concentrations in blood (B), minimal cholinesterase (ChE) activity in red blood cells (C), and plasma (D). Note: y-axes in (A) and (B) are on a logarithmic scale, and x-axes are on a logarithmic scale.

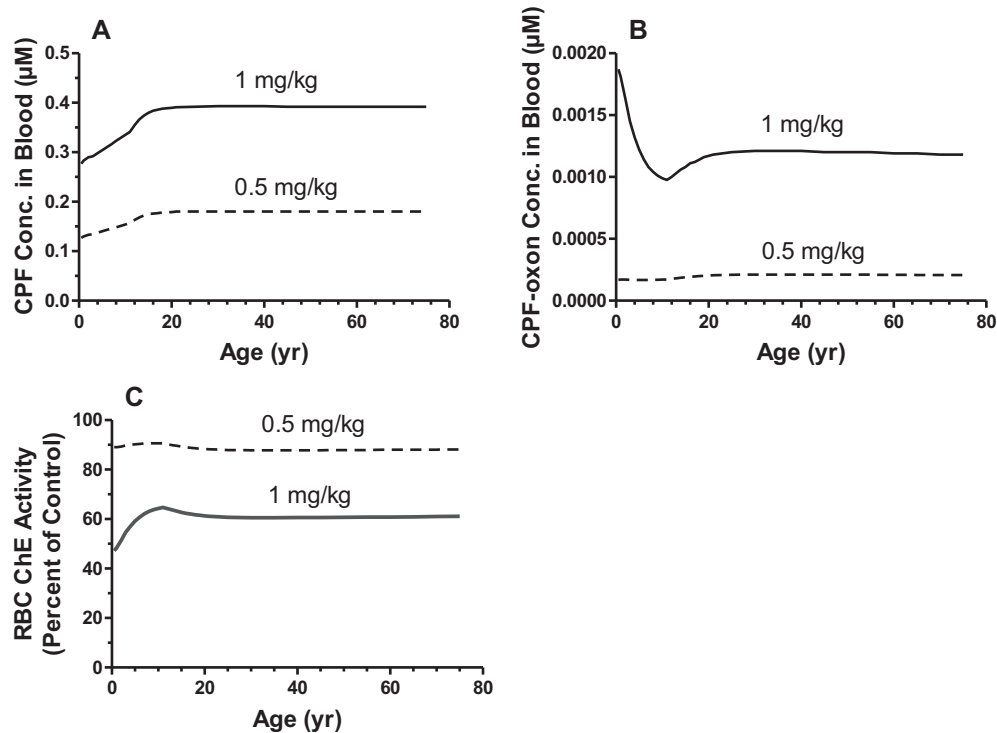


Fig. 10. Life-stage model predictions of humans dosed with 0.5 or 1 mg/kg chlorpyrifos (CPF) orally as a function of age. Predictions include peak CPF concentration in blood (A), chlorpyrifos-oxon (CPF-oxon) concentrations in blood (B), and minimum acetylcholinesterase (AChE) activity in red blood cells (RBC).

4. Discussion

A life-stage PBPK/PD model for chlorpyrifos was developed to predict chlorpyrifos disposition and biological response across

ages. Mathematical descriptions of age-dependent changes in body weight, organ volumes, and metabolism from literature sources and experimental studies were all incorporated into the model. Finally, model simulations were compared to controlled adult

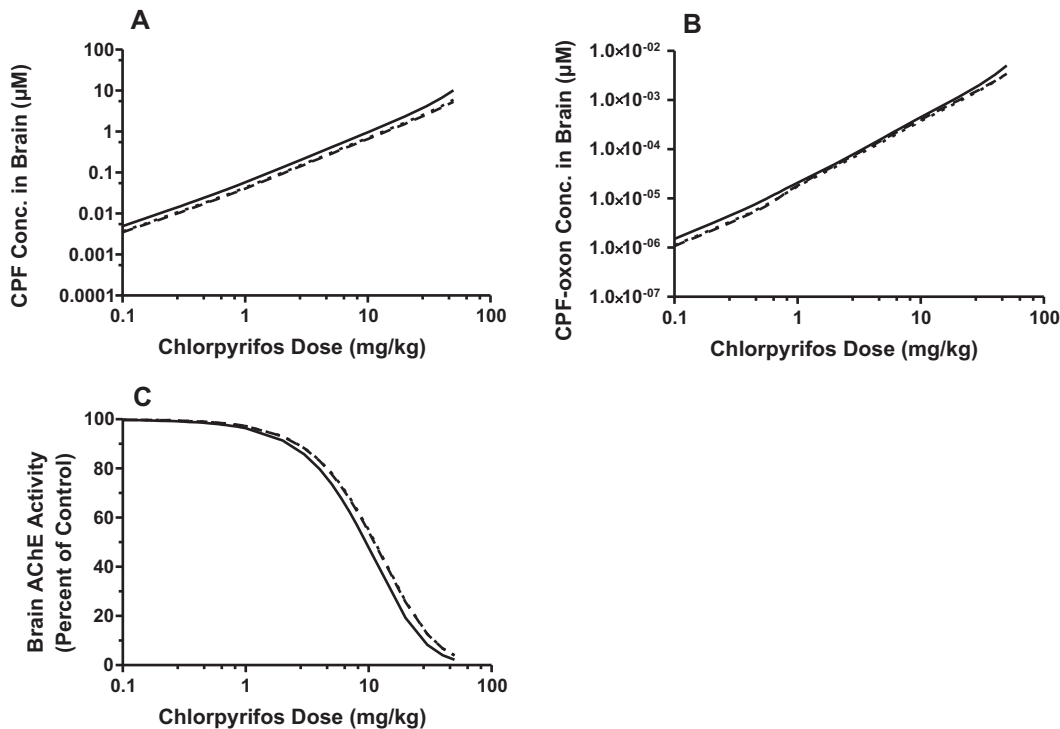


Fig. 11. Life-stage model predictions of 6 months infants (dashed line), 3 yr old children (dotted line), and 30 yr old adults (solid line) exposed to a variety of chlorpyrifos (CPF) doses. Predictions include peak CPF concentration in brain (A), chlorpyrifos-oxon (CPF-oxon) concentrations in brain (B), and minimum acetylcholinesterase (AChE) activity in brain. Note: y-axes in Fig. 10A and B are on a logarithmic scale, and x-axes are on a logarithmic scale.

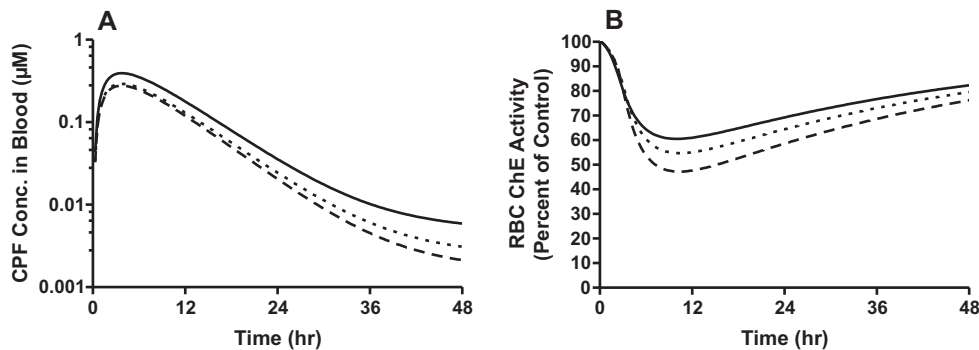


Fig. 12. Time course life-stage model predictions of 6 months infants (dashed line), 3 yr old children (dotted line), and 30 yr old adults (solid line) exposed to 1 mg/kg chlorpyrifos (CPF) dose. Predictions include CPF concentration in blood (A) and cholinesterase (ChE) activity in red blood cells (B). Note: y-axis in Fig. 11A is on a logarithmic scale.

Table 5
Area under the curve (AUC) and half-lives ($T_{1/2}$) of chlorpyrifos (CPF), chlorpyrifos-oxon (CPF-oxon), and 3,5,6-trichloro-2-pyridinol (TCPy) in blood as predicted for humans aged 0.5, 3, and 30 yr the life-stage model following an oral dose of 1 mg/kg chlorpyrifos.

Age (yr)	CPF		CPF-oxon		TCPy	
	AUC (0, 168) ($\mu\text{M} \times \text{h}$)	$T_{1/2}$ (h)	AUC (0, 168) ($\mu\text{M} \times \text{h}$)	$T_{1/2}$ (h)	AUC (0, 168) ($\mu\text{M} \times \text{h}$)	$T_{1/2}$ (h)
0.5	3.4	21	0.01	3.9	645	31
3	3.76	29	0.008	3.9	643	32
30	5.31	42	0.007	3.6	632	32

human exposure studies. Computational models provide a unique framework to quantitatively incorporate various age-dependent changes in physiology and mechanisms of chemical disposition to predict net effects on target tissue dosimetry and biological response.

Physiological values used in the PBPK/PD model were similar to what others use in standard PBPK models. Compartment volumes

obtained from literature sources and refit compartment volume parameters were very consistent with standard adult PBPK volume estimates (Fig. 3) (Brown et al., 1997) and other model estimates from our group including adults and 3 yr old children (Hinderliter et al., 2011). Overall, the sum of compartments within the model (total volume of distribution) accounted for ~75–95% of body weight depending on age not including skeletal bone and

Table 6

Sensitivity analysis for selected parameters of the life-stage model following an oral dose of 3 µg/kg chlorpyrifos (CPF) for 6 months and 30 yr old humans. Endpoints included acetylcholinesterase (AChE) inhibition in red blood cells and brain, and the sensitivity coefficients were calculated at the time of maximal inhibition. Categories of sensitivity include “Low” (absolute value of sensitivity coefficient < 0.1), “Med” (absolute value of sensitivity coefficient > 0.1 and < 0.5), and “High” (absolute value of sensitivity coefficient > 0.5).

Parameter	Sensitivity category			
	Red blood cell AChE Inhibition		Brain AChE Inhibition	
	6 months	30 yr	6 months	30 yr
<i>Metabolism</i>				
Hepatic dearylation	High	High	High	High
Hepatic desulfuration	High	High	Med	Low
Hepatic CPF-oxon hydrolysis	Low	High	High	High
Blood CPF-oxon hydrolysis	Low	High	Low	High
Brain dearylation	Low	Low	Med	Med
Brain desulfuration	Low	Low	High	High
<i>Compartment volume</i>				
Brain	Low	Low	High	High
Fat	Low	Low	Low	Low
Liver	Low	Low	High	High
<i>Blood flows</i>				
Brain	Low	Low	Low	Low
Liver	High	High	Med	Med
<i>Enzyme levels</i>				
Red blood cell AChE	High	High	Low	High
Brain AChE	Low	Low	High	High
Liver carboxylesterase	High	Med	High	High
Brain carboxylesterase	High	Low	High	High
<i>Fraction bound in plasma</i>				
CPF	High	High	High	High
CPF-oxon	High	High	High	High
<i>Partition coefficient</i>				
CPF				
Brain	Low	Low	High	High
Liver	Low	Low	High	High
Fat	Low	Low	Low	Med
<i>CPF-oxon</i>				
Brain	Low	Low	Low	Low
Liver	High	High	Low	Low
Fat	Low	Low	Low	Low

non-adipose connective tissue (Supplemental Fig. 4). Blood flows were also consistent with standard adult values and previously published values (Fig. 4) (Brown et al., 1997; Hinderliter et al., 2011), and the overall cardiac output (summation of blood flows) was very similar to more commonly used allometric scaling methods (Brown et al., 1997) (Supplemental Fig. 8). Using the algorithm approach to calculate the volume of distribution allowed for physiological changes from birth to adulthood to be accounted for in TCPy pharmacokinetics. Body weights are increasing with time (Ogden et al., 2004), and thus the use of cross sectional physiological data to simulate lifetime exposures may have some uncertainties, especially for longer simulation times.

There have been some concerns raised that chlorpyrifos-oxon hydrolysis previously measured *in vitro* (Smith et al., 2011) is not relevant for PBPK modeling, because measurements were conducted at non-physiological pH and salt concentration, and measurements did not account for both A- and B-esterase hydrolysis of chlorpyrifos-oxon (US EPA, 2011). Like many biological estimates experimentally derived *in vitro*, measurements of oxon hydrolysis have traditionally used experimental parameters (e.g. pH and salt concentrations) optimized for the *in vitro* system, which are not at standard physiological levels (Furlong et al., 1989; Richter and Furlong, 1999). The significance of using these *in vitro* estimates measured at non-physiological conditions, however, is a bit unclear. Hydrolysis rates of diazoxon, a similar organophosphate pesticide, demonstrated no differences for QQ (the most sensitive phenotype) and QR phenotypes incubated under optimized *in vitro* conditions and physiological conditions

(Richter et al., 2009). Diazoxon hydrolysis rates of RR phenotypes (least sensitive phenotype) were 16% slower in optimized *in vitro* conditions compared to physiological conditions (Richter et al., 2009). Additionally, chlorpyrifos-oxon hydrolysis in plasma was measured under physiological conditions, and activities levels ranged from 3 to 10 µmol/(min × mL) (Richter et al., 2009). Adult V_{max} values for chlorpyrifos-oxon hydrolysis in plasma used to parameterize the life-stage PBPK model were 4–9 µmol/(min × mL) (Smith et al., 2011), which is very similar to those measured previously under physiological conditions (Richter et al., 2009). Hepatic chlorpyrifos-oxon hydrolysis intrinsic clearance values used in the life-stage PBPK model (Smith et al., 2011) are consistent with previously reported values measured at physiological conditions (Sams et al., 2004). Using these metabolism estimates and optimized transfer rates, the model accurately predicts available human data (Figs. 5–8). Additional concerns have been raised regarding not controlling for chlorpyrifos-oxon hydrolysis by B-esterases while obtaining metabolism parameters for A-esterases *in vitro* (US EPA, 2011). Here, we demonstrate that B-esterases are completely inhibited by 500 nM of chlorpyrifos-oxon in human plasma (Supplemental Fig. 7). This indicates that the relative contribution of chlorpyrifos-oxon hydrolysis by B-esterases is insignificant at experimental conditions used to measure Michaelis-Menten metabolism kinetics of A-esterases (72.3–2400 µM chlorpyrifos-oxon in diluted [1/200] human plasma) (Smith et al., 2011), as 1×10^{-4} – $3 \times 10^{-3}\%$ of the chlorpyrifos-oxon would have inhibited B-esterases in plasma. Similarly in rat liver homogenates, it has been demonstrated that hepatic B-esterases are nearly

completely inhibited by 7.5 μM chlorpyrifos-oxon (Chanda et al., 1997; Mortensen et al., 1998a). Thus, we feel that available chlorpyrifos-oxon hydrolysis measurements are sufficient for parameterization of the life-stage model.

One difference between the previously published chlorpyrifos model in preweanling rats and this model was the scaling of B-esterase levels (acetyl-, butyryl-, and carboxyl-) (Timchalk et al., 2007b). Experimental evidence suggests that overall levels of acetyl- and butyryl-esterases increase with age in brain tissue of rats (Liu et al., 1999; Mortensen et al., 1998b; Tang et al., 1999) and were scaled allometrically by body weight in the rat neonate model (Timchalk et al., 2007b). However, these age-related differences were relatively insensitive when scaled by age-dependent brain volume, as simulations of 1 and 5 mg/kg chlorpyrifos caused an increase of brain cholinesterase inhibition by 1–3%, respectively, when using the scaled parameters versus an overall mean (data not shown). Due to the lack of sensitivity of these parameters and the lack of available human data (specifically age-dependent levels of B-esterases), these parameters were held constant and scaled by brain volume in the life-stage PBPK/PD model. Likewise, scaling cholinesterase activities by red blood cell and plasma volumes resulted in adequate age-dependent scaling of experimental values (Carr et al., 2001) and were handled as such in this life-stage model. Carboxylesterase activity has also demonstrated increased levels with age in the liver, plasma, and brain tissue of rats (Atterberry et al., 1997; Chanda et al., 2002; Karanth and Pope, 2000; Lassiter et al., 1999). However, due to limited human data, there is some uncertainty to whether this pattern persists across species. Pope et al. (2005) measured carboxylesterase activity in a limited number of human liver S9 fractions ages 2 months to 36 yr ($n = 10$) and found no statistical significant relationship, although samples from individuals 2 and 3 months old (the youngest samples tested) had the lowest activities. Ontogeny of human carboxylesterase expression has been measured in S9 liver fractions, and it was found that expression of human carboxylesterase-1 was significantly lower in children <1 yr of age, while human carboxylesterase-2 demonstrated no significant differences across age (Zhu et al., 2009). Yang et al. (2009) reported expression and activities for various drugs (e.g. aspirin, permethrin, etc.) for both human carboxylesterase-1 and -2 demonstrated the following pattern in pooled samples: fetuses < children < adults. The relationship between the relative expression of human carboxylesterase-1 and -2 with overall activity of carboxylesterase (measured with *p*-nitrophenyl acetate hydrolysis) is unknown. In humans, carboxylesterase is not found in plasma (Li et al., 2005), and no studies have investigated the ontogeny of carboxylesterase activity in brain tissue.

Other age-dependent factors were considered for incorporation into the model but were instead held constant levels of microsomal protein, overall levels of plasma proteins, and transfer rates. In rats, it has been demonstrated that microsomal protein levels increase from birth to postnatal day 7, where they approach adult levels (Alcorn et al., 2007), and in humans, a meta-analysis suggests some possibility of a similar trend existing in humans, although substantial operator-specific error and inter-individual variability exists (Barter et al., 2008). Because of that variability and small sample size of pediatric samples compared to adults, hepatic microsome levels are held constant in the life-stage PBPK/PD model. It was previously reported that levels of plasma proteins increase with age in humans (Alcorn and McNamara, 2003; Kanakoudi et al., 1995; Smith et al., 2011). This could have implications on levels of protein binding, especially at high exposures, as binding of chlorpyrifos and chlorpyrifos-oxon in plasma were sensitive parameters for both age groups in respect to brain acetylcholinesterase inhibition at the reference dose (Table 6) and red blood cell acetylcholinesterase inhibition at higher doses (data not shown).

However, since most risk-based model simulations focus on environmental exposures, it was assumed that possible age-dependent difference in plasma protein levels would have a negligible effect on binding of low levels of chlorpyrifos and chlorpyrifos-oxon in plasma, and fractional binding of chlorpyrifos and chlorpyrifos-oxon was held constant in this model. Age-dependent transfer rates with respect to chlorpyrifos pharmacokinetics have not been investigated, and thus they were held constant in this model.

The life-stage PBPK/PD model predicts similar responses in adults compared to previous PBPK/PD models for chlorpyrifos. Previously, a chlorpyrifos PBPK/PD model was published for adult rats and humans (Timchalk et al., 2002). This life-stage model built upon that model with several updates including partition coefficients, gut and brain metabolism, and age-dependent measures such as metabolism and anatomy. In comparison to the previous model, the life-stage model predicts very similar response in plasma cholinesterase inhibition up to 6 $\mu\text{g/kg/d}$; afterwards, the life-stage model predicts a more sensitive response (Fig. 13). Likewise, the life-stage model predicts more sensitive responses for acetylcholinesterase inhibition in red blood cells once significant inhibition occurs and a very similar response in brain acetylcholinesterase inhibition (Fig. 13). Parameter updates to the original model all contribute, to some degree, to these differences.

The life-stage model predicts age-dependent differences in chlorpyrifos pharmacokinetics and biological response. After equivalent oral doses of chlorpyrifos, the life-stage model predicts marginally lower systemic levels of chlorpyrifos and, at doses ≥ 0.6 mg/kg chlorpyrifos, higher levels of chlorpyrifos-oxon in children compared to adults. These age-dependent discrepancies resulted from differences in overall chlorpyrifos and chlorpyrifos-oxon metabolism and, to a lesser extent, chlorpyrifos distribution. Higher chlorpyrifos levels in adults resulted from lower levels of chlorpyrifos metabolism compared to children on a body weight basis as evidenced by a larger AUC of chlorpyrifos in blood and longer chlorpyrifos blood half-life (Table 5). Since levels of hepatic chlorpyrifos and chlorpyrifos-oxon metabolism on a microsomal basis and the level of hepatic microsomal protein are constant as a function of age in the model, increased hepatic chlorpyrifos and chlorpyrifos-oxon metabolism in children is driven by a larger liver fraction per body weight compared to adults. At doses ≥ 0.6 mg/kg (100- to 1000-fold higher than environmental exposure levels), predicted increases in chlorpyrifos-oxon in children are a result of chlorpyrifos-oxon levels overwhelming chlorpyrifos-oxon metabolism capacity in plasma, which is lower in children than in adults (Smith et al., 2011). At doses <0.6 mg/kg, increased chlorpyrifos-oxon metabolism in children is enough to cause marginally lower chlorpyrifos-oxon levels compared to adults. The model predicts that plasma cholinesterase inhibition is slightly higher in

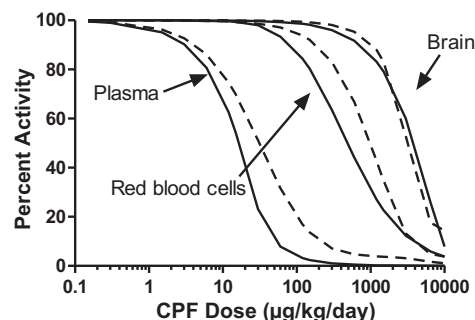


Fig. 13. Comparison of cholinesterase inhibition in plasma and acetylcholinesterase inhibition in red blood cells and brain between the life-stage physiologically based pharmacokinetic and pharmacodynamic (PBPK/PD) model (solid lines) and a previously published PBPK/PD model (dashed lines) for chlorpyrifos (CPF) in adults. Single doses were administered as zero order constant rates of uptake for both simulations.

adults than in children, because the dynamic range for cholinesterase inhibition in plasma occurs at doses lower than 0.6 mg/kg chlorpyrifos, and thus, chlorpyrifos-oxon levels are slightly higher in adults. These simulations (<0.6 mg/kg chlorpyrifos) are comparable to predictions made previously, suggesting that 19-year old humans are more sensitive to plasma and red blood cell cholinesterase inhibition than 1-year olds from equivalent oral doses of chlorpyrifos (Foxenberg et al., 2011).

Besides metabolism, distribution also plays a role in age-dependent differences in chlorpyrifos pharmacokinetics. Adults have more fat content than children, and since chlorpyrifos is lipophilic ($\log K_{ow}$ 4.82; McCall et al., 1980), fat can act as a chlorpyrifos depot. Following an oral dose of chlorpyrifos, adults have a larger fraction of the dose in fat depots compared to 6 months infants (~2-fold), which alters the distribution of chlorpyrifos. Lower overall chlorpyrifos metabolism and, at a lesser extent, altered distribution increases the half-life of chlorpyrifos in blood to nearly double in adults compared to 6 months old neonates (Table 5).

Dietary exposure is considered to be the primary route of non-occupational chlorpyrifos exposure (Lu et al., 2010). CARES and LifeLine are two dietary exposure models that have predicted CPF exposure levels (Price et al., 2011). CARES predicted 11 (2.7–47) and 3.4 (0.8–15) ng/kg/d median (95% confidence intervals) CPF exposure for children and adults, respectively; while LifeLine predicted 20 (4.7–67) and 5.7 (1.2–24) ng/kg/d for children and adults (Price et al., 2011). These levels are well below the threshold (0.6 mg/kg) where the life-stage model predicts children to be more sensitive than adults to chlorpyrifos from equivalent oral doses.

There is uncertainty regarding model simulations of the brain compartment. Recently, the potential importance of low-level brain metabolism has been suggested especially for exposure routes that lead to higher levels of chlorpyrifos in blood by circumventing first pass metabolism in oral exposures (Ellison et al., 2011; Smith et al., 2009). Here, high sensitivity coefficients for parameters describing chlorpyrifos desulfuration in brain calculated at a reference dose for all ages further highlight this issue. In the life-stage model, estimates of brain metabolism are derived from preliminary metabolism studies with rat brain tissue (Timchalk et al., 2012). Experimentally derived measures of carboxylesterase and PON1 activities in brain tissue are rare in adult humans and do not exist for infants or children. Thus with the lack of experimental data, there remains a high degree of uncertainty surrounding quantitative predictions of the brain compartment, and additional experimental focus on this area may be warranted.

A life-stage PBPK/PD model for chlorpyrifos in humans has been developed. The model was calibrated and validated against controlled adult human exposure studies. Simulations suggest age-dependent pharmacokinetics and response exist, where at doses ≥ 0.6 mg/kg, children have higher levels of chlorpyrifos-oxon in blood and higher levels of red blood cell cholinesterase inhibition from equivalent oral doses of chlorpyrifos. At doses <0.6, the model predicts that adults will have slightly higher levels of chlorpyrifos-oxon in blood, and thus inhibition of cholinesterase as well. It is important to note that these simulations are made at ages that the model has not been validated *in vivo*, as data from children exposed to pesticides are very rare and difficult to obtain. Until such data exists, authors feel that this model provides a strategy to make reasonable predictions regarding life-stage pharmacokinetics and response. In the future, this model will have variability of sensitive parameters incorporated into the model structure and linked to exposure models to predict a life-stage based population response to environmental levels of chlorpyrifos.

Conflict of interest

None.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.yrtph.2013.10.005>.

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